

Decision 4 – Rejected Transaction Defaults: Stata comparison

Verification of the COOPECON app (branch migration, HEAD f8dff1e) against the professor's Decision 4 design document "Decision 4 - Rejected Transaction Defaults 040926-2.docx" and the Stata file

Stata_File_Decision4_050926.dta (280 participants). Prepared 2026-10-07 by the migration controller; every figure pair was checked bar by bar (bar count, bin edges and bar heights of the app chart against the histogram of the .dta column at the document's bin count; discrete charts against the value counts) – see the per-figure table in Appendix A.

How to read this document

Each step shows, side by side, the Stata figure (or Stata output) and the corresponding app screen or value, states the exact settings that produce the pair, and ends with a verdict line. The Stata figures are the ones embedded in the professor's document (with all tracked changes accepted; the previous revision's figures that the tracked changes deleted are not used). Stata itself could not be run (the StataNow licence on this machine expired on 30 Aug 2026), so every Stata summary number quoted here (Obs / Mean / SD / Min / Max, counts) was computed from the columns stored in the .dta with pandas, exactly as "summarize" / "tab" would report them. App charts are exported from the exact Plotly figure objects of the running app (Chrome, Plotly.toImage); app screens are Chrome captures; app numbers come from the same seam/engine path the "Run" buttons execute.

Settings used for every comparison (Stata parameter values)

Setting	Value	Why
Page 1 › Population Mode	Research Baseline	the original 280 participants, no stochastic component (Stata: the 280 observations). Both research modes use the 280 participants in file order: agent k = participant $k \bmod 280$ ($N = 280 \rightarrow$ exactly the file; $N = 1000 \rightarrow$ three full cycles plus the first 160; $N < 280 \rightarrow$ the first N); no bootstrap or permutation (owner ruling, HEAD e0d3d97)
Page 1 › Number of Agents	280	the Stata file has 280 observations; with the default 1000 the first 160 participants would be counted four times and the min-max segment statistics would shift slightly; seed irrelevant in the research modes (42 used)
Page 1 › Income	professor's income file (research modes)	$z_net_income = \text{std}(\text{income})$ over the 280, as in the .dta
Page 2 › Decision 4 › Income Specification	Continuous only / Categorical only	WTP and Risk-Taking are the only income-using elements
Page 2 › Decision 4 › Intercepts $\beta_0 \dots \beta_4$	all 0.0000 (= app defaults, config/decisions.yaml)	the .dta embeds no intercept; the app default β_0 (Options List Length, applied on the raw composite, override range $-0.25 \dots 0.25$) is 0 (owner ruling 2026-10-07), so the Stata comparison runs at the app defaults
Page 2 › Decision 4 › Flexibility Anchor Mix W_OFlex	0.25 (default)	$\text{anchored_flexibility} = 0.25 \cdot z_stdactions + 0.75 \cdot z_Flexibility_calculated_ivw$

Page 2 › Decision 4 › Stochastic	off (Research Baseline is always deterministic)	σ tables compared separately (step 10); a Research Specification run with σ on is shown in step 12
Histogram bins	Stata default rule $k = \text{trunc}(\min(\sqrt{N}, 10 \cdot \log_{10} N)) \rightarrow 16$ bins at $N = 280$	verified on the document's figures: every one of the 11 continuous histograms reproduces <code>np.histogram(.dta column, bins = 16)</code> bar for bar (max relative bar error $\leq 0.5\%$; 17 bins give 19–40 % errors). The app applies the same truncating rule (capped at 16; HEAD e41968d – until then it rounded to 17) and, since HEAD 0c46d6d, plots DENSITY (count / ($N \times$ bin width)) like Stata, so the side-by-side charts have the same bars, edges and heights

Summary of verdicts

Step	Stata item	App item	Verdict
1	Standardisation (egen std of the Big-5, education, stdactions)	YAML z_scoring vs .dta summarize	Match (7 significant digits)
2	Element equations and coefficients (TTP, Loyalty, WTP, Risk-Taking, Flexibility)	doc equations vs app sub-tabs	Match
3	F1 histogram weighted_ttp, F2 choice_length_deterministic	app "Tendency to Plan score" / "Number of pre-selected options"	Match – 280/280 agents ($\beta_0 = 0$); figures identical bar by bar
4	F3 histogram weighted_loyalty, F4 weighted_loyalty15	app "Loyalty score" / loyalty-based options ranking	Match – 280/280 (score, segment, 5-option list); figures identical bar by bar
5	F6 histogram z_WTP_calculated, F8 WTP_calculated15 (continuous)	app WTP score / ranking, Continuous only	Match – 280/280; figures identical bar by bar
6	T1 regress WTP (categorical), F5/F7 categorical WTP histograms	app quintile-effects table / WTP score, Categorical only	Coefficients match; F7 DOES NOT MATCH the app chart (max bar difference 25 % of the tallest bar): the .dta's stored weighted_WTP_categorical was generated with a superseded formula (see step 6) – the app reproduces the document's printed code; 1 agent's segment differs from the .dta
7	F9 histogram z_RT_calculated_hs, F12 RT_calculated15	app Risk-Taking score / ranking, Continuous only	Match – 280/280; figures identical bar by bar
8	T2 regress RT (categorical), F10/F11 categorical RT histograms	app quintile-effects table / RT score, Categorical only	Match – 280/280 (z-score and floor-rule segments); F11 identical bar by bar
9	F13 z_Flexibility_calculated_iww, F14 z_stdactions, F15 z_anchored_flexibility, F16 Flexibility_combined15	app Flexibility score / ranking + export columns	Match – 280/280; F15/F16 identical bar by bar (F13/F14 have no app chart: columns 280/280)
10	σ tables (TTP, Loyalty, WTP, Risk-Taking, Flexibility)	app "Base σ and effective σ per budget level"	Match the document literally (two arithmetic slips inside the document itself are noted)

11	Section 6 integrated default list (Kemeny-Young + tie-break)	app "Integrated Default List" + tie statistics	App-only (no Stata counterpart); output rules hold 280/280
12	Stochastic component (no Stata figure)	Research Specification run with σ on	App-only example; UI = engine (identical allocation)
13	Compare both / Compare all layouts	app comparison screens	Consistent with the single-mode runs; Copula differs by construction

Step 1: Settings in the app

Page 1 (population mode and agents) and the Decision 4 tab (income specification, stochastic component, intercept override):

Deploy ⓘ

COOPECON AI Agent Simulation

Page 1: Common Simulation Parameters

Simulation Execution Mode

Execution Mode ⓘ
☒ Snapshot ☐ Live Simulation

Simulation Mode

Analysis Mode ⓘ
☒ Single Run ☐ Monte-Carlo Study

Single Run: Execute one simulation with current parameters

Simulation Settings

Number of Agents ⓘ

280 - +

Random Seed ⓘ

Time Parameters

Market Parameters

Platform Markup (m) ⓘ
 0.10

Price Range (r) ⓘ
 0.25

Bidding Percentage (bp) ⓘ
 0.50

Products for auction per vendor: 50

Price Grid Categories (g) ⓘ
 11 - +

Annual Income Distribution

Research Baseline and Research Specification use each original participant's income from the professor's data file; these distribution settings apply to Copula populations. ⓘ

Deploy

Apply Purchasing Limits?

☐ Yes

☒ No

Population Mode

☐ Copula (synthetic)

☐ Research Specification

☒ Research Baseline

☐ Compare all

Purchasing limits are disabled. For the practical purpose of running the simulation, an artificial purchasing limit is specified here, indicating a maximum number of purchased items for all agents based on their assumed purchasing preferences and budget limit.

Note: This artificial limit represents a market-based constraint derived from customer budget preferences and needs, rather than a regulatory purchasing restriction.

Max Purchases per Term (All Agents)

50

Reset all simulation parameters (vendors, prices, distributions) to system defaults.

Reset to Default Values

Next: Decision Parameters →

Enhanced AI Agent Simulation Framework | Two-Page Interface

Deploy

COOPECON AI Agent Simulation

Page 2: Decision-Specific Parameters

Decision Selection

☐ Select All Decisions

Select Decisions to Run

4. Rejected Trans... x

Overview

4. Rejected Transaction Defaults

Rejected Transaction Defaults Configuration

Income Specification

Income Specification for Rejected Transaction Model

☐ Categorical only

☒ Continuous only

☐ Compare both

Stochastic Component

Research Baseline always uses deterministic scores. Configure stochastic settings for Copula / Research Specification runs.

Copula Mode:

☐ Add Normal(anchor, σ) draw to Copula runs

Research Specification Mode:

☒ Use Normal(anchor, σ) draw in Research Specification mode

Options List Length sub-tab: intercept override 0 (Research Default 0.0000, the app default):

Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates each agent's **Tendency to Plan** based on Big-5 personality traits and education, and then converts it into the number of pre-selected default options (0-5).

$$TendencyToPlan_i (TTP_i) = \beta_0 - 0.015255656 \times z_{Extraversion_i} + 0.01776386 \times z_{Agreeableness_i} + 0.01959 \times z_{Neuroticism_i} + 0.00901465 \times z_{Conscientiousness_i} + 0.0297 \times Edu_i$$

$$OptionsListLength05_i = \left\lfloor (6 - 0.0001) \times \frac{TTP_i - \min(TTP)}{\max(TTP) - \min(TTP)} \right\rfloor \in \{0, \dots, 5\}$$

Stochastic Component:

If stochastic enabled: $TTP_i \sim \text{Normal}(\mu = \text{anchor}, \sigma)$ where the anchor is the continuous TTP_i score and $\sigma = \text{base } \sigma \times \text{coefficient}$ (overall or per budget level); the drawn values are re-scaled over the population and re-binned into the 0-5 options list length.

Intercept Override

Research Default (β_0): 0.0000

Override Value

Baseline Options List Length tendency ⓘ

0,0000

Change

Step 2: The standardisation of values

Stata commands (document, section 5 and the per-element "Variable Standardisation" blocks):

```
egen z_extraversionbig5 = std(extraversionbig5)
egen z_agreeable = std(agreeable)
egen z_neuroticismbig5 = std(neuroticismbig5)
egen z_conscientiousnessbig5 = std(conscientiousnessbig5)
egen z_opennessbig5 = std(opennessbig5)
egen z_net_income = std(income)
gen rededucation = education-1
egen z_stdactions = std(stdactions)
```

The app freezes the original-280 means and SDs in config/decisions.yaml (rejected_transaction_defaults.z_scoring) and uses the professor's income file for the research modes. "summarize" of the raw .dta columns vs the app constants:

Variable	Obs	Stata mean	Stata SD	App mean	App SD	Equal
extraversionbig5	280	3.557857	0.6989565	3.557857	0.6989565	yes
agreeable	280	3.546071	0.3732712	3.546071	0.3732712	yes
neuroticismbig5	280	2.702143	0.6839657	2.702143	0.6839657	yes
conscientiousnessbig5	280	3.657143	0.5596521	3.657143	0.5596521	yes
opennessbig5	280	4.060714	0.5068274	4.060714	0.5068274	yes
stdactions	280	1.186338	0.6752843	1.186338	0.6752843	yes
income (z_net_income)	280	28334.95	17711.6	computed per run from the professor's income file		yes (engine income stats mean 28334.95, sd 17711.60)
rededucation	280	0.3785714	0.4858995	Education – 1 per agent		yes

The values match exactly (to the 7 significant digits Stata prints).

Step 2b: The element equations

The document's final equations and the app's sub-tab formulas, coefficient by coefficient:

Element	Document	App (config/decisions.yaml, shown on the sub-tabs)	Equal
TTP (doc §1)	$\beta_0 - 0.0152556564 \cdot zE + 0.0177638642 \cdot zA + 0.01959 \cdot zN + 0.00901465 \cdot zC + 0.0297 \cdot \text{reducation}$	$-0.0152556564 / 0.0177638642 / 0.01959 / 0.00901465 / 0.0297$	yes
Loyalty (doc §2, bs_weighted_loyalty)	$\beta_1 - 0.009828468 \cdot zE + 0.01096706 \cdot zO + 0.0123046 \cdot zA$ (then egen std)	$-0.009828468 / 0.01096706 / 0.0123046$	yes
WTP (doc §3, H&S)	$\beta_2 + 0.078863062 \cdot zE - 0.012326128 \cdot zA + 0.698 \cdot z_net_income$	$0.078863062 / -0.012326128 / 0.698$	yes
Risk-Taking (doc §4, H&S)	$\beta_3 + 0.025942386297 \cdot zE + 0.023699214948 \cdot zO - 0.038734315188 \cdot zA - 0.037739440732 \cdot zC - 0.025388697852 \cdot zN + 0.006874197106 \cdot z_net_income$	$0.025942386297 / 0.023699214948 / -0.038734315188 / -0.037739440732 / -0.025388697852 / 0.006874197106$	yes
Flexibility (doc §5, IVW)	$\beta_4 + 0.0206 \cdot zE + 0.0294118 \cdot zO - 0.04921357 \cdot zN + 0.04339814 \cdot zA + 0.04811179 \cdot zC$; anchored = $0.25 \cdot z_stdactions + 0.75 \cdot z_calc$	$0.0206 / 0.0294118 / -0.04921357 / 0.04339814 / 0.04811179$; W_OFlex 0.25	yes
Priority sequences	Loyalty 3>1>4>5>2; WTP 3>2>1>4>5; RT 4>2>1>3>5; Flexibility 2>4>3>1>5	loyalty 3>1>4>5>2; wtp 3>2>1>4>5; risk_taking 4>2>1>3>5; flexibility 2>4>3>1>5	yes

Deploy ⓘ

Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) [2. Loyalty Ranking](#) 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Loyalty to the vendor based on Big-5 personality traits.

$$Loyalty_i = \beta_1 - 0.009828468 \times z_{Extroversion_i} + 0.01096706 \times z_{Openness_i} + 0.0123046 \times z_{Agreeableness_i}$$

$$Loyalty15_i = \left\lceil 1 + (5 - 0.0001) \times \frac{Loyalty_i - \min(Loyalty)}{\max(Loyalty) - \min(Loyalty)} \right\rceil \in \{1, \dots, 5\}$$

Loyalty rejected transaction options sequence: Option 3 > Option 1 > Option 4 > Option 5 > Option 2

Loyalty score segment	Priority list for rejected transaction options	Options list length
1 (Lowest 20% of Loyalty score segment)	2	1
2	5 > 2	2
3	4 > 5 > 2	3
4	1 > 4 > 5 > 2	4
5 (Highest 20% of Loyalty score segment)	3 > 1 > 4 > 5 > 2	5

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

Option 3: current vendor at PN price

Option 4: place a bid

Option 5: forgo the transaction

Stochastic Component:

If stochastic enabled: $Loyalty_i \sim \text{Normal}(\mu = \text{anchor}, \sigma)$ where the anchor is the continuous $Loyalty_i$ score and $\sigma = \text{base } \sigma \times \text{coefficient}$ (overall or per budget level); the drawn values are re-scaled over the available options based on the 15-point sequence.

Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. **Willingness-to-Pay Ranking** 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Willingness to Pay based on personality traits and income.

$$WTP_i = \beta_2 + 0.078863062 \times z_{Extroversion_i} - 0.012326128 \times z_{Agreeableness_i} + 0.698 \times z_{Income_i}$$

$$WTP15_i = \left\lfloor 1 + (5 - 0.0001) \times \frac{WTP_i - \min(WTP)}{\max(WTP) - \min(WTP)} \right\rfloor \in \{1, \dots, 5\}$$

Willingness-to-Pay rejected transaction options sequence: Option 3 > Option 2 > Option 1 > Option 4 > Option 5

Willingness-to-Pay score segment	Priority list for rejected transaction options	Options list length
1 (Lowest 20% of Willingness-to-Pay score segment)	5	1
2	4 > 5	2
3	1 > 4 > 5	3
4	2 > 1 > 4 > 5	4
5 (Highest 20% of Willingness-to-Pay score segment)	3 > 2 > 1 > 4 > 5	5

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

Option 3: current vendor at PN price

Option 4: place a bid

Option 5: forgo the transaction

Probabilistic Parameters:

Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. **Risk-Taking Ranking** 5. Flexibility Ranking 6. Integrated Default List

Estimates Risk-Taking propensity based on Big-5 personality traits and income.

$$RiskTaking_i = \beta_3 + 0.0259423863 \times z_{Extroversion_i} + 0.0236992149 \times z_{Openness_i} - 0.0387343152 \times z_{Agreeableness_i} - 0.0377394407 \times z_{Conscientiousness_i} - 0.0253886979 \times z_{Neuroticism_i}$$

$$RiskTaking15_i = \left\lfloor 1 + (5 - 0.0001) \times \frac{RiskTaking_i - \min(RiskTaking)}{\max(RiskTaking) - \min(RiskTaking)} \right\rfloor \in \{1, \dots, 5\}$$

Risk-Taking rejected transaction options sequence: Option 4 > Option 2 > Option 1 > Option 3 > Option 5

Risk-Taking score segment	Priority list for rejected transaction options	Options list length
1 (Lowest 20% of Risk-Taking score segment)	5	1
2	3 > 5	2
3	1 > 3 > 5	3
4	2 > 1 > 3 > 5	4
5 (Highest 20% of Risk-Taking score segment)	4 > 2 > 1 > 3 > 5	5

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

Option 3: current vendor at PN price

Option 4: place a bid

Option 5: forgo the transaction

Probabilistic Parameters:

Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Flexibility based on Big-5 personality traits and observed flexibility in the experiment (captured by standard deviation in the number of actions across the eight cycle-weeks).

$$\text{CalculatedFlexibility}_i = \beta_4 + 0.0206 \times z_{\text{Extraversion}_i} + 0.0294118 \times z_{\text{Openness}_i} - 0.04921357 \times z_{\text{Neuroticism}_i} + 0.04339814 \times z_{\text{Agreeableness}_i} + 0.04811179 \times z_{\text{Conscientiousness}_i}$$

$$\text{AnchoredFlexibility}_i = W_{\text{Flex}} \times \text{ObservedFlexibility}_i + W_{\text{CFlex}} \times \text{CalculatedFlexibility}_i = 0.25 \times \text{ObservedFlexibility}_i + 0.75 \times \text{CalculatedFlexibility}_i$$

ObservedFlexibility is the observed flexibility variable stdactions (the standard deviation in the number of actions across the eight cycle-weeks): a copula trait for synthetic populations and the participant's own value in the research baseline and research specification modes. All variables are standardized prior to calculation.

$$\text{Flexibility15}_i = \left\lceil 1 + (5 - 0.0001) \times \frac{\text{AnchoredFlexibility}_i - \min(\text{AnchoredFlexibility})}{\max(\text{AnchoredFlexibility}) - \min(\text{AnchoredFlexibility})} \right\rceil \in \{1, \dots, 5\}$$

Flexibility rejected transaction options sequence: Option 2 > Option 4 > Option 3 > Option 1 > Option 5

Flexibility score segment	Priority list for rejected transaction options	Options list length
1 (Lowest 20% of Flexibility score segment)	5	1
2	1 > 5	2
3	3 > 1 > 5	3
4	4 > 3 > 1 > 5	4
5 (Highest 20% of Flexibility score segment)	2 > 4 > 3 > 1 > 5	5

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

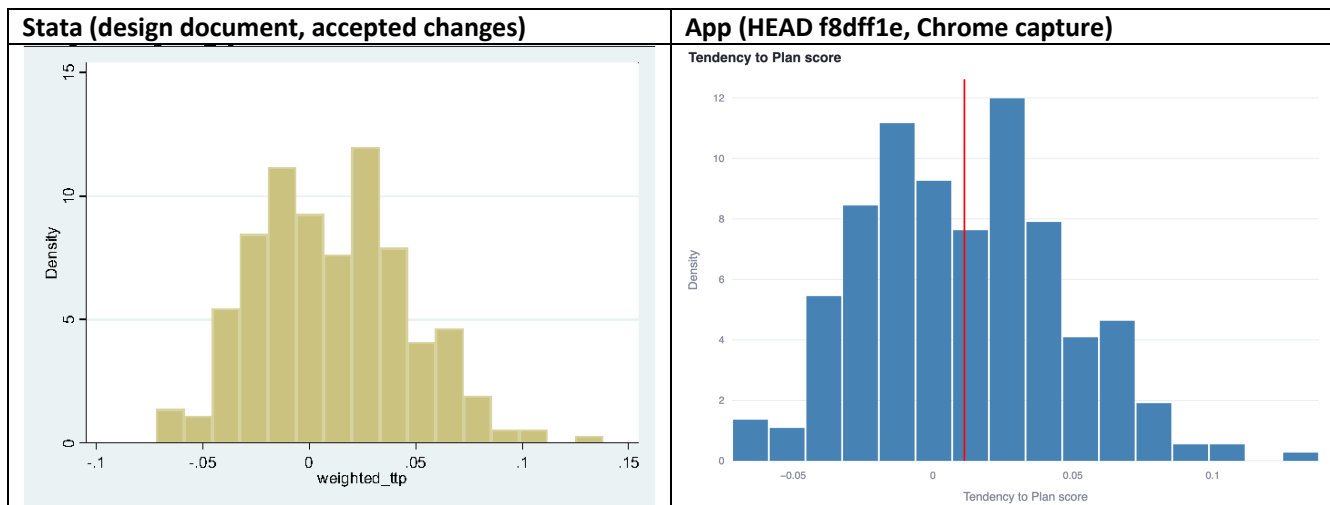
The equations and every coefficient match the document.

Step 3: Options List Length (Tendency to Plan) – F1, F2

Settings: Research Baseline, 280 agents, $\beta_0 = 0$ (the app default since HEAD ffa32fb; a non-zero β_0 would shift the raw score and, with the app's fixed-cutoff semantics, the allocation; the Stata file has no intercept). Income specification irrelevant (no income term).

```
gen weighted_ttp = (-0.0152556564 * z_extraversionbig5) + (0.0177638642 * z_agreeable) + (0.01959
* z_neuroticismbig5) + (0.00901465 * z_conscientiousnessbig5) + (0.0297 * reduction)
summarize weighted_ttp
histogram weighted_ttp
```

Variable	Stat a Obs	Stata Mean	Stata SD	Stata Min	Stata Max	Ap p Ob s	App Mean	App SD	App Min	App Max
weighted_ttp	280	0.01124357	0.03528479	-0.07178839	0.1379994	280	0.01124358	0.03528479	-0.07178839	0.1379994



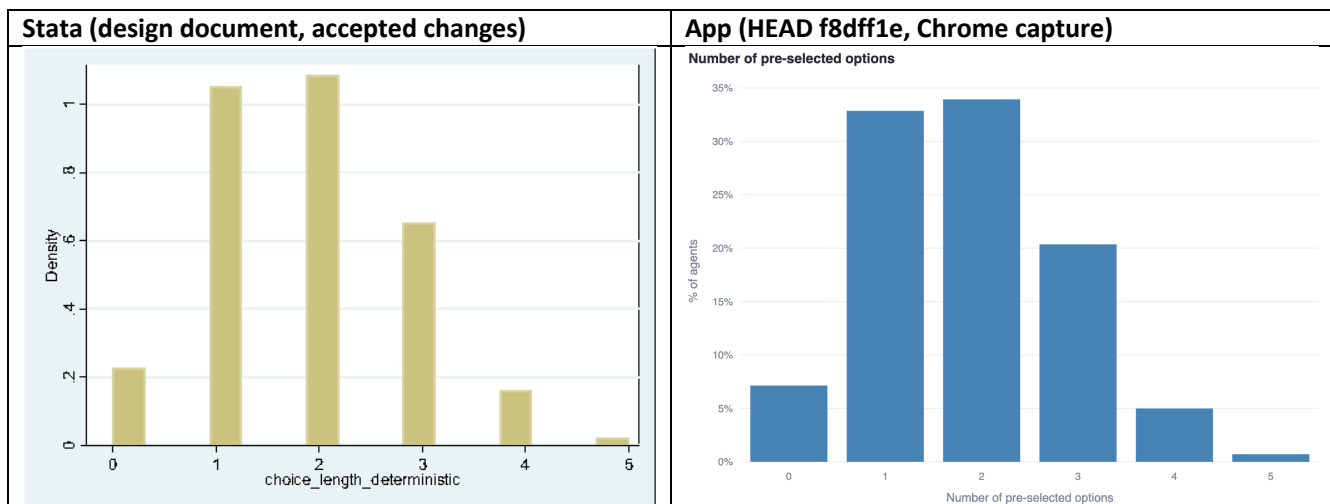
```

gen weighted_ttp06 = (6-0.0001)*((weighted_ttp - min_weighted_ttp)/(max_weighted_ttp -
min_weighted_ttp))
gen choice_length_deterministic = floor(weighted_ttp06)

```

Variable	Stata Obs	Stata Mean	Stata SD	Stata Min	Stata Max	App Obs	App Mean	App SD	App Min	App Max
choice_length_deterministic	280	1.853571	1.031748	0	5	280	1.853571	1.031748	0	5

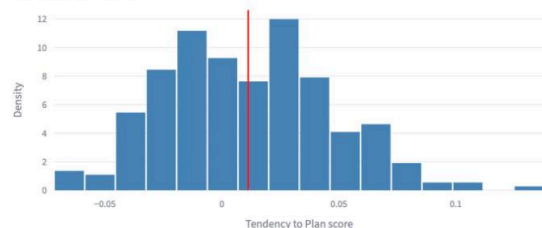
Category	Stata count	Stata %	App count	App %	Equal
options 0	20	7.1%	20	7.1%	yes
options 1	92	32.9%	92	32.9%	yes
options 2	95	33.9%	95	33.9%	yes
options 3	57	20.4%	57	20.4%	yes
options 4	14	5.0%	14	5.0%	yes
options 5	2	0.7%	2	0.7%	yes



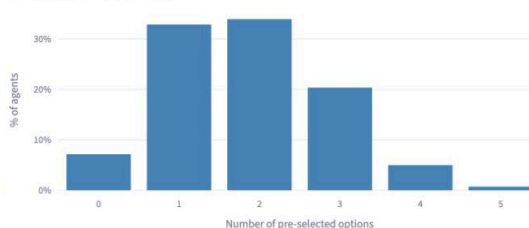
presents how many default options each agent pre-selects (0-5)

Min -0.0718 · Max 0.1380

Tendency to Plan score



Number of pre-selected options



Number of pre-selected options	% of agents
0	7.1%
1	32.9%
2	33.9%
3	20.4%
4	5.0%
5	0.7%

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

The values match exactly: `weighted_ttp` agrees for 280/280 participants (max |diff| 1.8e-08) and `choice_length_deterministic` for 280/280. Figures bar by bar – F1: 16 bars vs 16 in the Stata figure, bin edges equal to 7.7e-09, max bar-height difference 3.6e-08 density units (0.000 % of the tallest bar); F2: 6 bars vs 6, max bar-height difference 0.000 percentage points. Both charts plot the same quantity as Stata (density / % of agents).

Step 4: Loyalty ranking – F3, F4

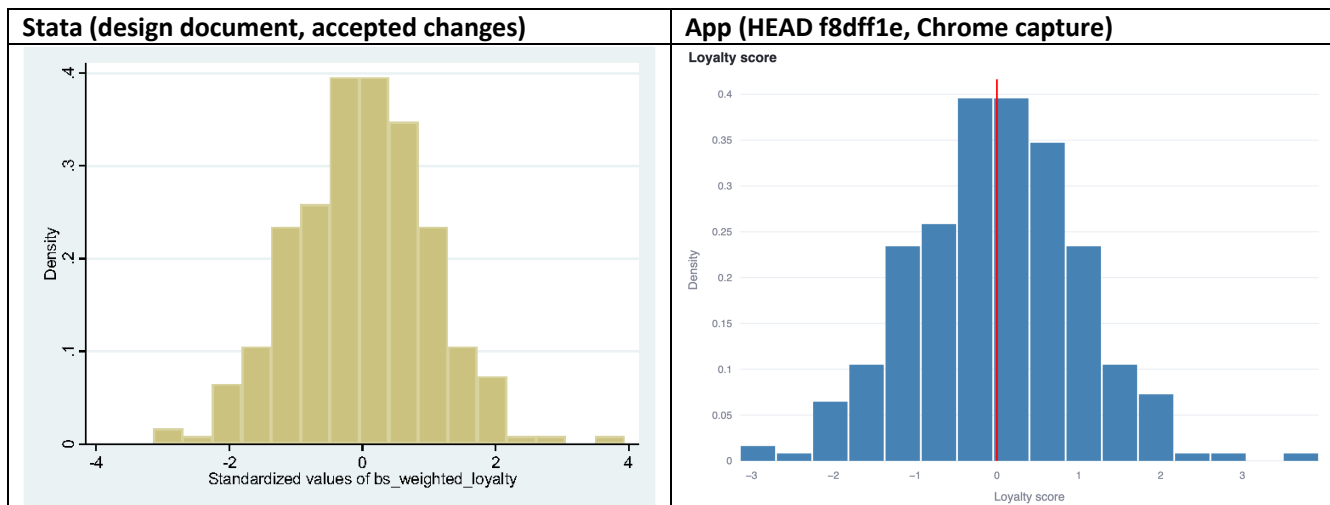
Settings as in step 3 ($\beta_1 = 0$; no income term).

```

gen bs_weighted_loyalty = -0.009828468*z_extraversionbig5 + 0.01096706*z_opennessbig5 +
0.0123046*z_agreeable
egen weighted_loyalty = std(bs_weighted_loyalty)
replace weighted_loyalty = weighted_loyalty + beta1
summarize weighted_loyalty
histogram weighted_loyalty

```

Variable	Stata Obs	Stata Mean	Stata SD	Stata Min	Stata Max	App Obs	App Mean	App SD	App Min	App Max
weighted_loyalty	280	-1.250633e-09	1	-3.141984	3.93632	280	-3.806479e-17	1	-3.141984	3.93632



```

gen weighted_loyalty15 = floor(1+(5-0.0001)*((weighted_loyalty -
min_weighted_loyalty)/(max_weighted_loyalty - min_weighted_loyalty)))
summarize weighted_loyalty15
histogram weighted_loyalty15
replace choicel_loyalty_deterministic = 2 if weighted_loyalty15 == 1 ... = 3 if
weighted_loyalty15 == 5

```

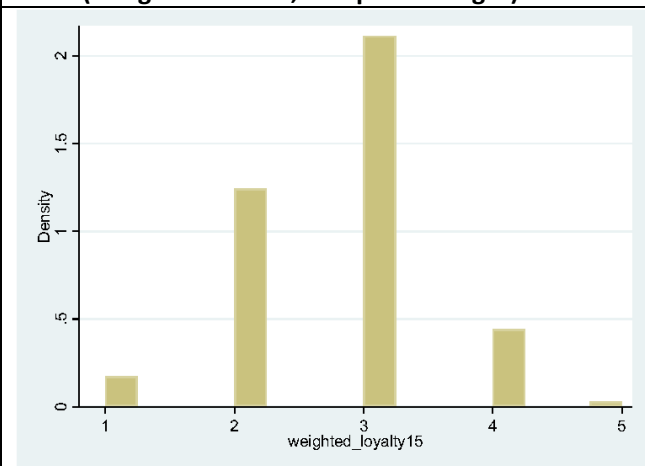
The app chart shows the FIRST option of each agent's loyalty list, with the x-axis in reversed priority order (2, 5, 4, 1, 3) – i.e. exactly segments 1...5 of weighted_loyalty15 relabelled with their first option:

Variable	Stata Obs	Stata Mean	Stata SD	Stata Min	Stata Max	App Obs	App Mean	App SD	App Min	App Max
weighted_loyalty15	280	2.728571	0.74143	1	5	280	2.728571	0.74143	1	5

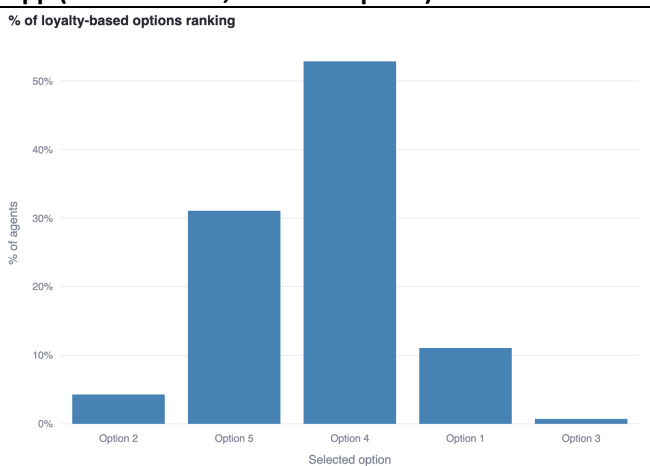
Category	Stata count	Stata %	App count	App %	Equal
segment 1	12	4.3%	12	4.3%	yes
segment 2	87	31.1%	87	31.1%	yes
segment 3	148	52.9%	148	52.9%	yes
segment 4	31	11.1%	31	11.1%	yes
segment 5	2	0.7%	2	0.7%	yes

Category	Stata count	Stata %	App count	App %	Equal
first option 1	31	11.1%	31	11.1%	yes
first option 2	12	4.3%	12	4.3%	yes
first option 3	2	0.7%	2	0.7%	yes
first option 4	148	52.9%	148	52.9%	yes
first option 5	87	31.1%	87	31.1%	yes

Stata (design document, accepted changes)



App (HEAD f8dff1e, Chrome capture)



Deploy

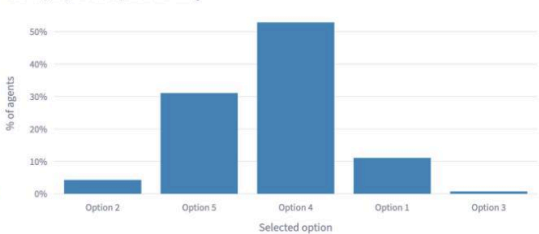
Priority sequence Option 3 > Option 1 > Option 4 > Option 5 > Option 2

Min -3.1420 · Max 3.9363

Loyalty score



% of loyalty-based options ranking



Option	% of agents
Option 1	11.1%
Option 2	4.3%
Option 3	0.7%
Option 4	52.9%
Option 5	31.1%

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

Option 3: current vendor at PN price

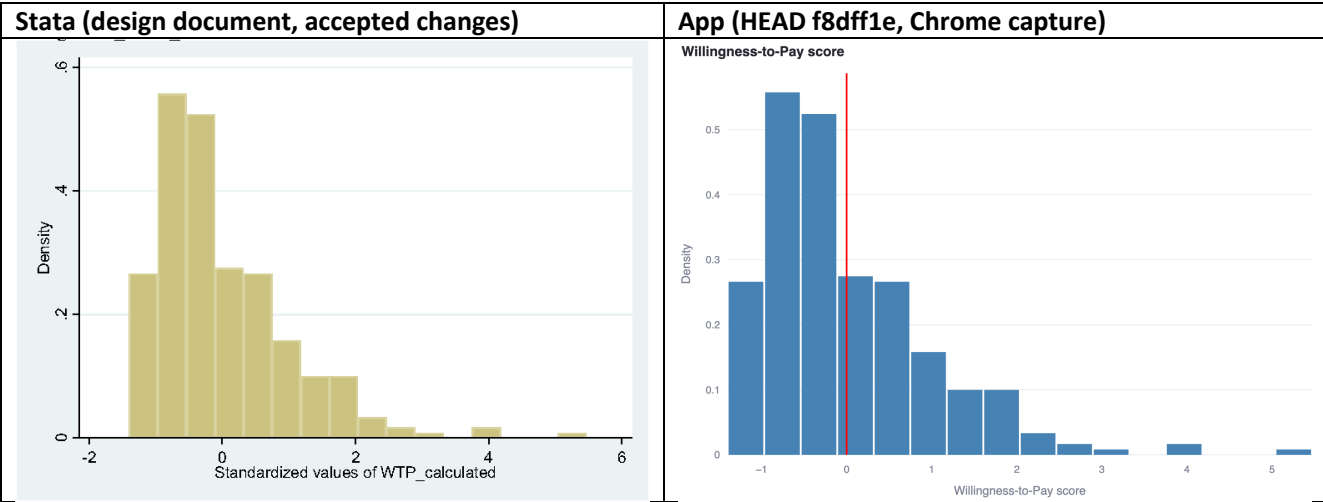
The values match exactly: weighted_loyalty 280/280 (max |diff| 3.2e-07), weighted_loyalty15 280/280, and the five-position choice lists choice1...choice5_loyalty_deterministic 280/280. Figures bar by bar – F3: 16 bars vs 16 in the Stata figure, bin edges equal to 7.2e-08, max bar-height difference 2.6e-09 density units (0.000 % of the tallest bar); F4 (segments 1...5 ↔ first options 2, 5, 4, 1, 3): 5 bars vs 5, max bar-height difference 0.000 percentage points.

Step 5: Willingness-to-Pay ranking, continuous income – F6, F8

Settings as in step 3 with Income Specification = Continuous only (β2 = 0).

```
gen WTP_calculated = 0.078863062 * z_extraversionbig5 - 0.012326128* z_agreeable + 0.698 * z_net_income
egen z_WTP_calculated= std(WTP_calculated)
replace z_WTP_calculated = z_WTP_calculated + beta2
summarize z_WTP_calculated
histogram z_WTP_calculated
```

Variable	Stata Obs	Stata Mean	Stata SD	Stata Min	Stata Max	App Obs	App Mean	App SD	App Min	App Max
z_WTP_calculated	280	1.736251e-09	1	-1.397873	5.470183	280	-1.268826e-17	1	-1.397877	5.470187

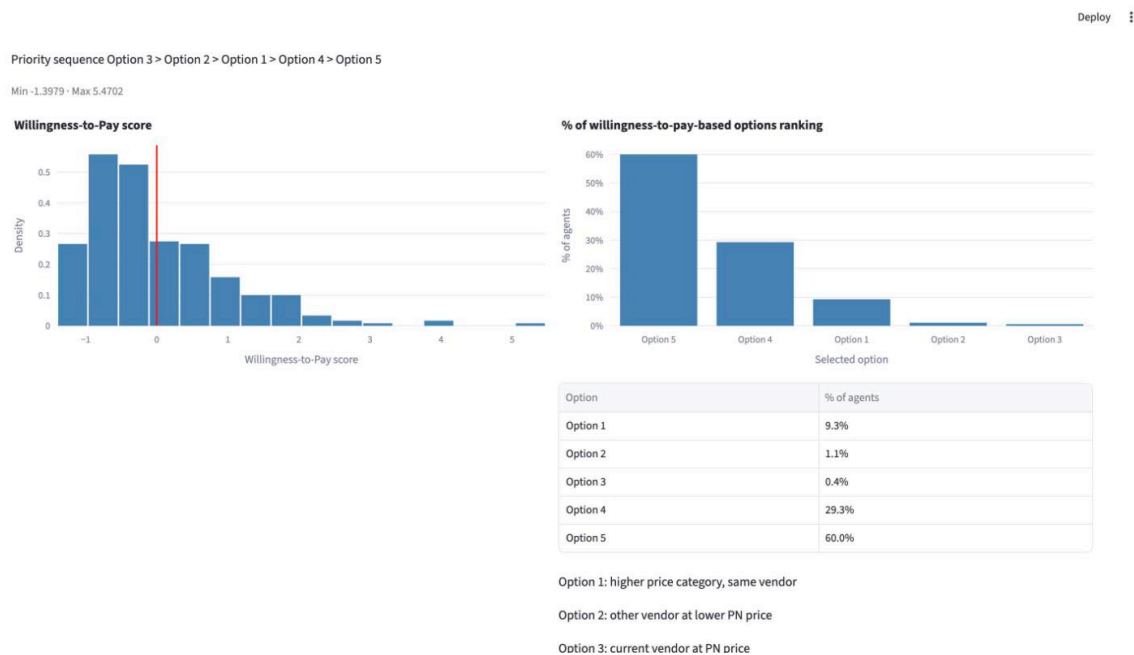
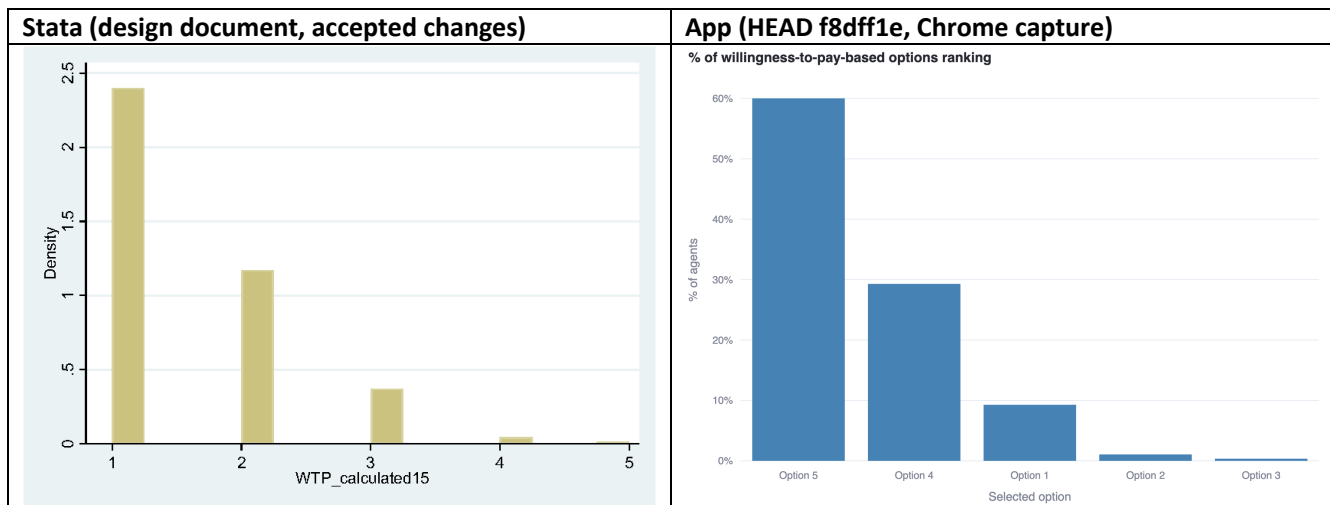


```
gen WTP_calculated15 = floor(1+(5-0.0001)*(( z_WTP_calculated - min_WTP_calculated)/(
max_WTP_calculated - min_WTP_calculated)))
histogram WTP_calculated15
```

Variable	Stata Obs	Stata Mean	Stata SD	Stata Min	Stata Max	App Obs	App Mean	App SD	App Min	App Max
WTP_calculated15	280	1.525	0.7376826	1	5	280	1.525	0.7376826	1	5

Category	Stata count	Stata %	App count	App %	Equal
segment 1	168	60.0%	168	60.0%	yes
segment 2	82	29.3%	82	29.3%	yes
segment 3	26	9.3%	26	9.3%	yes
segment 4	3	1.1%	3	1.1%	yes
segment 5	1	0.4%	1	0.4%	yes

Category	Stata count	Stata %	App count	App %	Equal
first option 1	26	9.3%	26	9.3%	yes
first option 2	3	1.1%	3	1.1%	yes
first option 3	1	0.4%	1	0.4%	yes
first option 4	82	29.3%	82	29.3%	yes
first option 5	168	60.0%	168	60.0%	yes



The values match exactly: `z_WTP_calculated` 280/280 (max |diff| 8.1e-06, the .dta stores float32), `WTP_calculated15` 280/280, choice lists 280/280. Figures bar by bar – F6: 16 bars vs 16 in the Stata figure, bin edges equal to 3.9e-06, max bar-height difference 5.8e-07 density units (0.000 % of the tallest bar); F8: 5 bars vs 5, max bar-height difference 0.000 percentage points.

Step 6: Willingness-to-Pay ranking, categorical income – T1, F5, F7

Settings as in step 3 with Income Specification = Categorical only ($\beta_2 = 0$).

```
regress WTP_calculated WTP_noincome i.totalallowance
      WTP_noincome .5439353 | 32 .2671588 | 72 .5057716 | 128 .9411747 | 200 1.822471 | _cons -
      .691843
gen weighted_WTP_categorical = cond(assignedallowancelevel ==5, 1.822471-0.691843 + 0.078863062 *
z_extraversionbig5 - 0.012326128 * z_agreeable, cond(... ==4, 0.9411747-0.691843 + ..., cond(...
==3, 0.5057716-0.691843 + ..., cond(... ==2, 0.2671588-0.691843 + ..., -0.691843 + 0.078863062 *
z_extraversionbig5 - 0.012326128 * z_agreeable)))
```

```
egen z_WTP_categorical = std(weighted_WTP_categorical)
replace z_WTP_categorical = z_WTP_categorical+ beta2
```

The regression table's level effects appear in the app as the "Income Quintile Effects (β_{income_q})" table (base intercept + level dummy):

Quintile	Stata (regress, document)	App β_{income_q}	Equal
Q1 (€12)	-.691843	-0.6918430	yes
Q2 (€32)	-.691843 + .2671588 = -.4246842	-0.4246842	yes
Q3 (€72)	-.691843 + .5057716 = -.1860714	-0.1860714	yes
Q4 (€128)	-.691843 + .9411747 = .2493317	0.2493317	yes
Q5 (€200)	-.691843 + 1.822471 = 1.130628	1.1306280	yes

Deploy ⓘ

Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Willingness to Pay based on personality traits and income.

$$WTP_i = \beta_2 + 0.078863062 \times z_{Estroversion_i} - 0.012326128 \times z_{Agreeableness_i} + \beta_{income_q}[quintile_i]$$

Income Quintile Effects (β_{income_q}):

Quintile	β_{income_q}
Q1 (€12)	-0.6918430
Q2 (€32)	-0.4246842
Q3 (€72)	-0.1860714
Q4 (€128)	0.2493317
Q5 (€200)	1.1306280

β_{income_q} : income quintile effects based on agent's income category (Quintiles 1-5)

Each value = base intercept (-0.6918430) + the quintile's differential income effect (Quintile 1 is the base level).

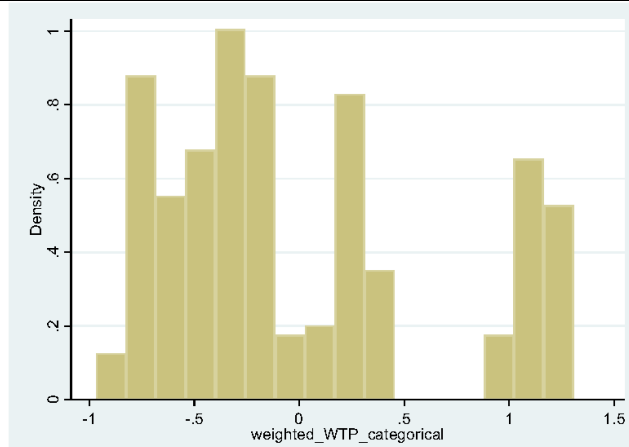
$$WTP15_i = \left\lceil 1 + (5 - 0.0001) \times \frac{WTP_i - \min(WTP)}{\max(WTP) - \min(WTP)} \right\rceil \in \{1, \dots, 5\}$$

Willingness-to-Pay rejected transaction options sequence: Option 3 > Option 2 > Option 1 > Option 4 > Option 5

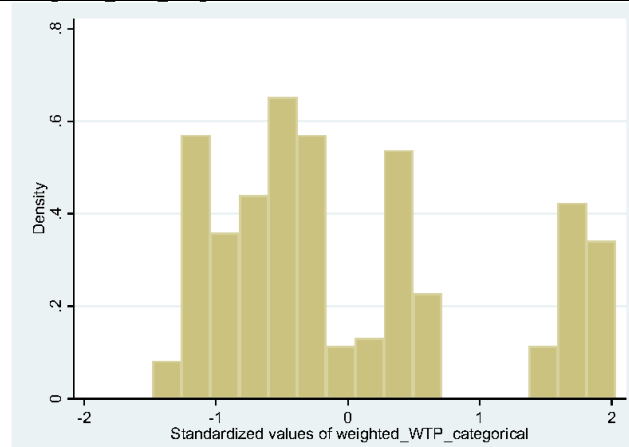
Willingness-to-Pay score segment	Priority list for rejected transaction options	Options list length
1 (Lowest 20% of Willingness-to-Pay score cammanfi)	5	1

Stata figures F5 (raw weighted_WTP_categorical) and F7 (standardised) vs the app's categorical WTP score (the app charts the standardised score; F5 is the same histogram on the raw scale):

Stata F5 – raw weighted_WTP_categorical



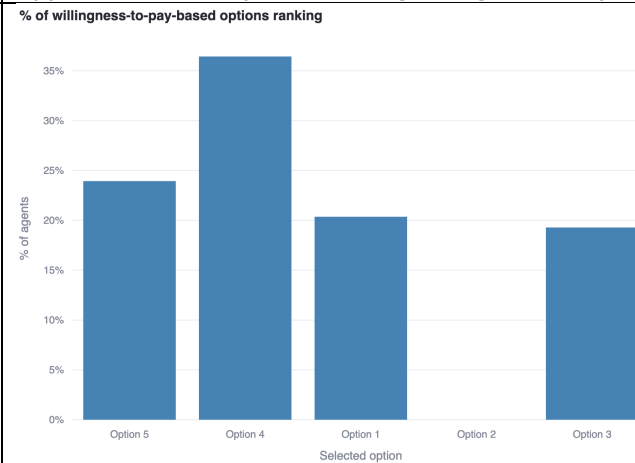
Stata F7 – z_WTP_categorical



App – WTP score, Categorical only



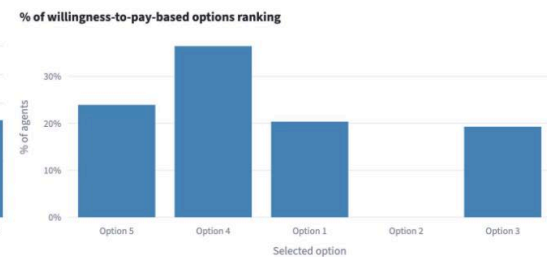
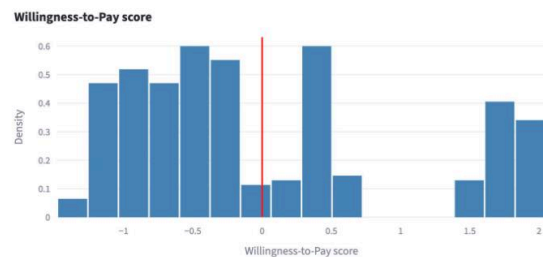
App – WTP-based options ranking, Categorical only



Deploy

Priority sequence Option 3 > Option 2 > Option 1 > Option 4 > Option 5

Min -1.4779 · Max 2.0467



Option	% of agents
Option 1	20.4%
Option 2	0.0%
Option 3	19.3%
Option 4	36.4%
Option 5	23.9%

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

Option 3: current vendor at PN price

Variable	Stat a Obs	Stata Mean	Stata SD	Stata Min	Stata Max	Ap p Obs	App Mean	App SD	App Min	App Max
z_WTP_categorical (.dta)	280	-2.328306e-10	1	-1.480752	2.029977	280	3.172066e-18	1	-1.477873	2.04668
weighted_WTP_categorical (.dta, raw)	280	-0.00869533	0.647716	-0.9677956	1.306144	280	-4.3748e-08	0.6381788	-0.9431473	1.306148

Finding (document / .dta error, not an app defect). The app evaluates the document's printed cond() code exactly (max |diff| 9e-09 against a direct re-computation of that code on the .dta z-columns), but the column weighted_WTP_categorical STORED in Stata_File_Decision4_050926.dta was generated with a superseded version of that command: for allowance levels 2–5 it uses the agreeableness coefficient -0.0123242128 (the value the tracked changes of rev 040926-2 deleted in favour of -0.012326128), and for level 1 (57 participants) a different intercept and coefficient altogether (fitted exactly: $-0.7355455 + 0.078863062 \cdot zE - 0.0046002694 \cdot zA$, instead of $-0.691843 + 0.078863062 \cdot zE - 0.012326128 \cdot zA$). Consequently the .dta's z_WTP_categorical differs from the document's own formula by up to 0.062 (z-units), its min/max are -1.480752 / 2.029977 versus -1.477873 / 2.046680 for the printed code, and the F7 histogram above is the histogram of the superseded column. The segment allocation differs for exactly 1 of 280 participants (Stata-stored column: 67/101/58/0/54; document code = app: 67/102/57/0/54). The app follows the written document; the professor may want to regenerate the .dta column with the current cond() command.

Category	Stata count	Stata %	App count	App %	Equal
segment (Stata-stored column vs app) 1	67	23.9%	67	23.9%	yes
segment (Stata-stored column vs app) 2	101	36.1%	102	36.4%	NO
segment (Stata-stored column vs app) 3	58	20.7%	57	20.4%	NO
segment (Stata-stored column vs app) 4	0	0.0%	0	0.0%	yes
segment (Stata-stored column vs app) 5	54	19.3%	54	19.3%	yes

DOES NOT MATCH visually: Stata F7 vs the app's categorical WTP score – 16 bars vs 16 in the Stata figure, bin edges equal to 1.7e-02, max bar-height difference 1.6e-01 density units (24.686 % of the tallest bar). Reason: F7 is the histogram of the .dta's stored z_WTP_categorical, which was generated with the superseded cond() formula described above (different level-1 intercept/coefficient and the old -0.0123242128 agreeableness weight), so its bar heights differ from the histogram of the document's printed code, which the app reproduces to 1e-08. The derived segment chart differs for exactly 1 of 280 agents (5 bars vs 5, max bar-height difference 0.357 percentage points). The coefficients and the app's computation match the document exactly; the stored .dta column is the only mismatch.

Step 7: Risk-Taking ranking, continuous income – F9, F12

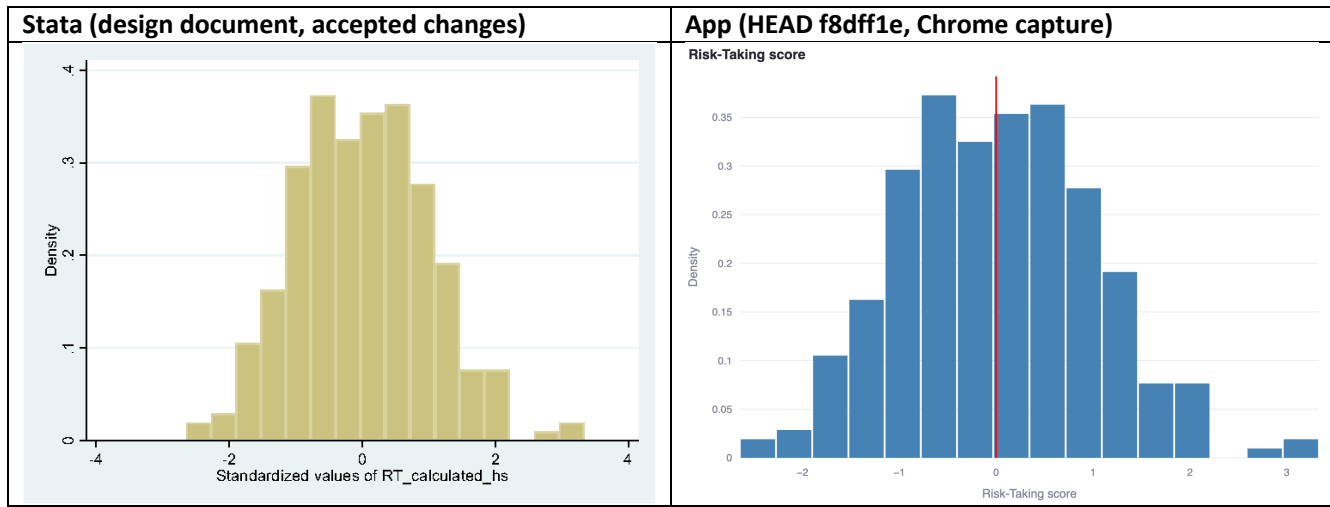
Settings as in step 5 ($\beta_3 = 0$, Continuous only).

```

gen RT_calculated_hs = 0.025942386297 * z_extraversionbig5 + 0.023699214948 * z_opennessbig5 -
0.038734315188 * z_agreeable - 0.037739440732 * z_conscientiousnessbig5 - 0.025388697852 *
z_neuroticismbig5 + 0.006874197106 * z_net_income
egen z_RT_calculated_hs = std(RT_calculated_hs)
replace z_RT_calculated_hs = z_RT_calculated_hs + beta3
histogram z_RT_calculated_hs

```

Variable	Stat a Obs	Stata Mean	Stat a SD	Stata Min	Stata Max	Ap p Obs	App Mean	Ap p SD	App Min	App Max
z_RT_calculated_h s	280	-8.421401e-1 0	1	-2.64659 4	3.33315 3	280	-2.854859e-1 7	1	-2.64659 4	3.33315 3



```

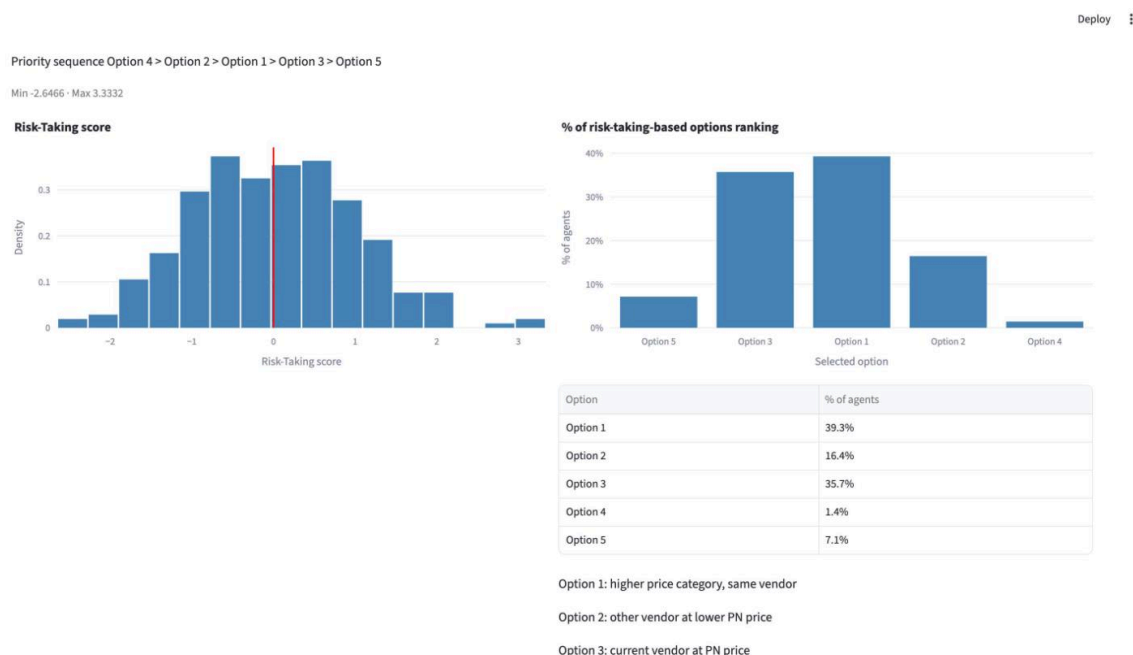
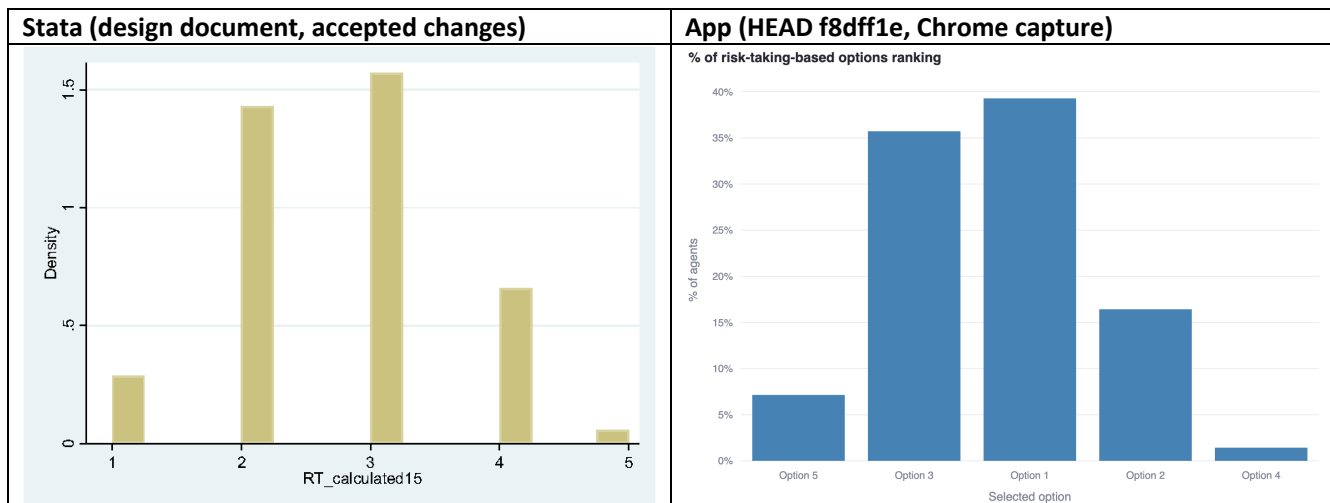
gen RT_calculated15 = floor(1+(5-0.0001)*((z_RT_calculated_hs -
min_RT_calculated)/(max_RT_calculated - min_RT_calculated)))
histogram RT_calculated15

```

Variable	Stata Obs	Stata Mean	Stata SD	Stata Min	Stata Max	App Obs	App Mean	App SD	App Min	App Max
RT_calculated15	280	2.692857	0.8790385	1	5	280	2.692857	0.8790385	1	5

Category	Stata count	Stata %	App count	App %	Equal
segment 1	20	7.1%	20	7.1%	yes
segment 2	100	35.7%	100	35.7%	yes
segment 3	110	39.3%	110	39.3%	yes
segment 4	46	16.4%	46	16.4%	yes
segment 5	4	1.4%	4	1.4%	yes

Category	Stata count	Stata %	App count	App %	Equal
first option 1	110	39.3%	110	39.3%	yes
first option 2	46	16.4%	46	16.4%	yes
first option 3	100	35.7%	100	35.7%	yes
first option 4	4	1.4%	4	1.4%	yes
first option 5	20	7.1%	20	7.1%	yes



The values match exactly: `z_RT_calculated_hs` 280/280 (max |diff| 1.7e-07), `RT_calculated15` 280/280, choice lists 280/280. Figures bar by bar – F9: 16 bars vs 16 in the Stata figure, bin edges equal to 7.4e-08, max bar-height difference 1.3e-09 density units (0.000 % of the tallest bar); F12: 5 bars vs 5, max bar-height difference 0.000 percentage points.

Step 8: Risk-Taking ranking, categorical income – T2, F10, F11

Settings as in step 6 ($\beta_3 = 0$, Categorical only).

```
regress RT_calculated_hs RT_noincome_hs i.totalallowance
      32 .0026128 | 72 .0050555 | 128 .0092738 | 200 .0179812 | _cons -.0068307
gen weighted_RT_categorical = cond(assignedallowancelevel ==5, 0.0179812-0.0068307 + ...,
cond(... ==4, 0.0092738-0.0068307 + ..., cond(... ==3, 0.0050555-0.0068307 + ..., cond(... ==2,
0.0026128-0.0068307 + ..., -0.0068307 + ...)))
egen z_RT_categorical = std(weighted_RT_categorical)
replace z_RT_categorical = z_RT_categorical+ beta3
```

Quintile	Stata (regress, document)	App $\beta_{_income_q}$	Equal
Q1 (€12)	-.0068307	-0.0068307	yes
Q2 (€32)	-.0068307 + .0026128 = -.0042179	-0.0042179	yes
Q3 (€72)	-.0068307 + .0050555 = -.0017752	-0.0017752	yes
Q4 (€128)	-.0068307 + .0092738 = .0024431	0.0024431	yes
Q5 (€200)	-.0068307 + .0179812 = .0111505	0.0111505	yes

Deploy ⓘ

Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Risk-Taking propensity based on Big-5 personality traits and income.

$$RiskTaking_i = \beta_3 + 0.0259423863 \times z_{Extroversion_i} + 0.0236992149 \times z_{Openness_i} - 0.0387343152 \times z_{Agreeableness_i} - 0.0377394407 \times z_{Conscientiousness_i} - 0.0253886979 \times z_{Neuroticism_i}$$

Income Quintile Effects ($\beta_{_income_q}$):

Quintile	$\beta_{_income_q}$
Q1 (€12)	-0.0068307
Q2 (€32)	-0.0042179
Q3 (€72)	-0.0017752
Q4 (€128)	0.0024431
Q5 (€200)	0.0111505

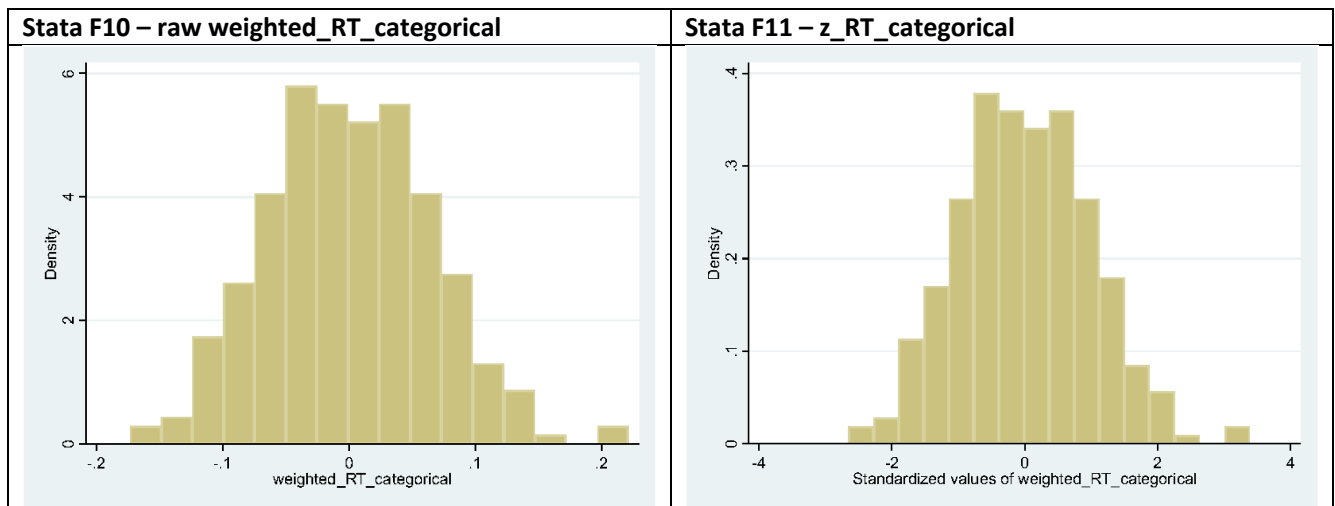
$\beta_{_income_q}$: Income quintile effects based on agent's income category (Quintiles 1-5)

Each value = base intercept (-0.0068307) + the quintile's differential income effect (Quintile 1 is the base level).

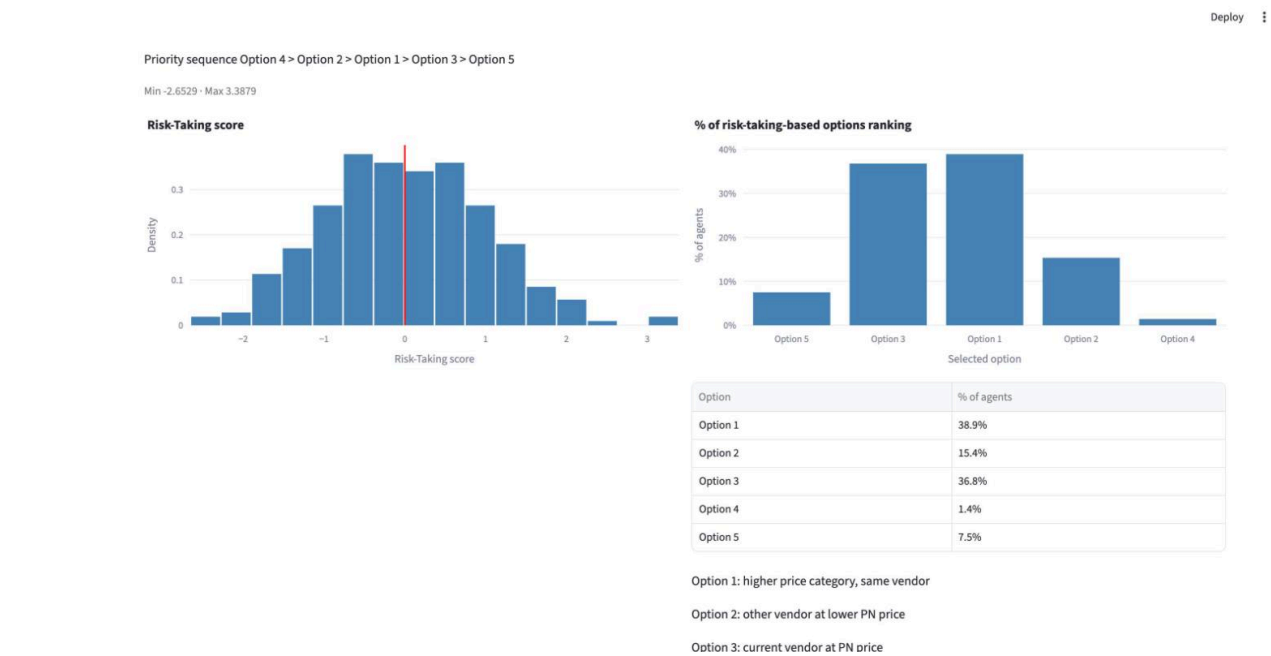
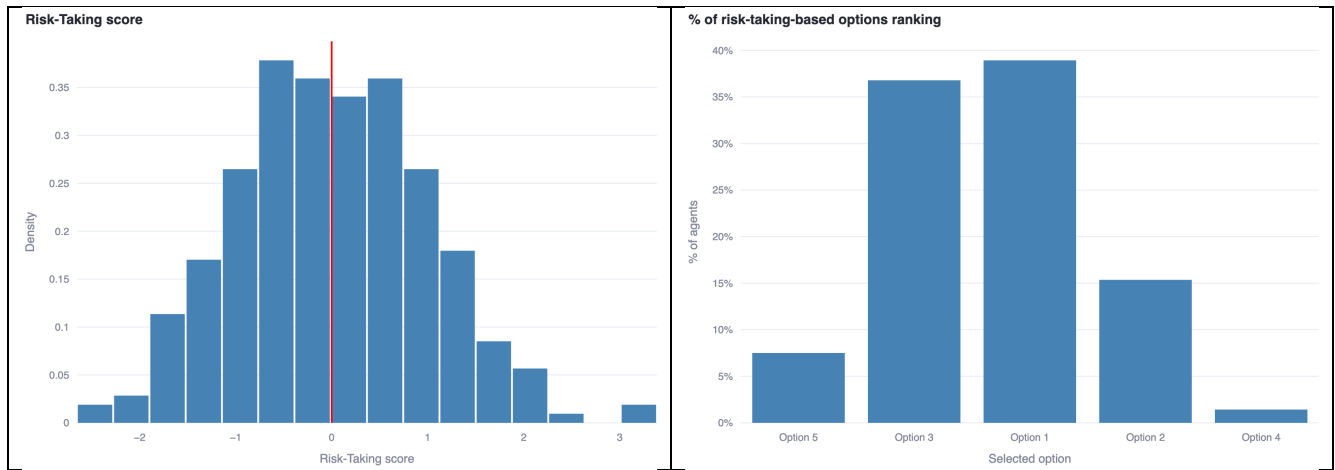
$$RiskTaking15_i = \left\lceil 1 + (5 - 0.0001) \times \frac{RiskTaking_i - \min(RiskTaking)}{\max(RiskTaking) - \min(RiskTaking)} \right\rceil \in \{1, \dots, 5\}$$

Risk-Taking rejected transaction options sequence: Option 4 > Option 2 > Option 1 > Option 3 > Option 5

Risk-Taking score segment	Priority list for rejected transaction options	Options list length
1 (Lowest 20% of Risk-Taking score command1)	5	1



App – Risk-Taking score, Categorical only	App – RT-based options ranking, Categorical only
---	--



Variable	Stat a Obs	Stata Mean	Stata SD	Stata Min	Stata Max	Ap p Obs	App Mean	App SD	App Min	App Max
z_RT_categorical	280	8.53158e-10	1	-2.652918	3.387871	280	9.516197e-18	1	-2.652918	3.387871
weighted_RT_categorical (raw)	280	-4.175324e-08	0.06530878	-0.1732589	0.2212577	280	-5.230291e-08	0.06530878	-0.1732589	0.2212577

The document stores no categorical RT segments; applying its floor rule to the .dta's z_RT_categorical gives 21/103/109/43/4, identical to the app's categorical RT segments.

Category	Stata count	Stata %	App count	App %	Equal
segment (floor rule on .dta vs app) 1	21	7.5%	21	7.5%	yes
segment (floor rule on .dta vs app) 2	103	36.8%	103	36.8%	yes

segment (floor rule on .dta vs app) 3	109	38.9%	109	38.9%	yes
segment (floor rule on .dta vs app) 4	43	15.4%	43	15.4%	yes
segment (floor rule on .dta vs app) 5	4	1.4%	4	1.4%	yes

The values match exactly: z_RT_categorical 280/280 (max |diff| 2.0e-07); floor-rule segments 280/280. Figures bar by bar – F11 vs the app's categorical Risk-Taking score: 16 bars vs 16 in the Stata figure, bin edges equal to 1.5e-07, max bar-height difference 1.3e-08 density units (0.000 % of the tallest bar); floor-rule segments of the .dta column vs the app ranking chart: 5 bars vs 5, max bar-height difference 0.000 percentage points. (F10 is the same histogram on the raw scale; the app charts the standardised score.)

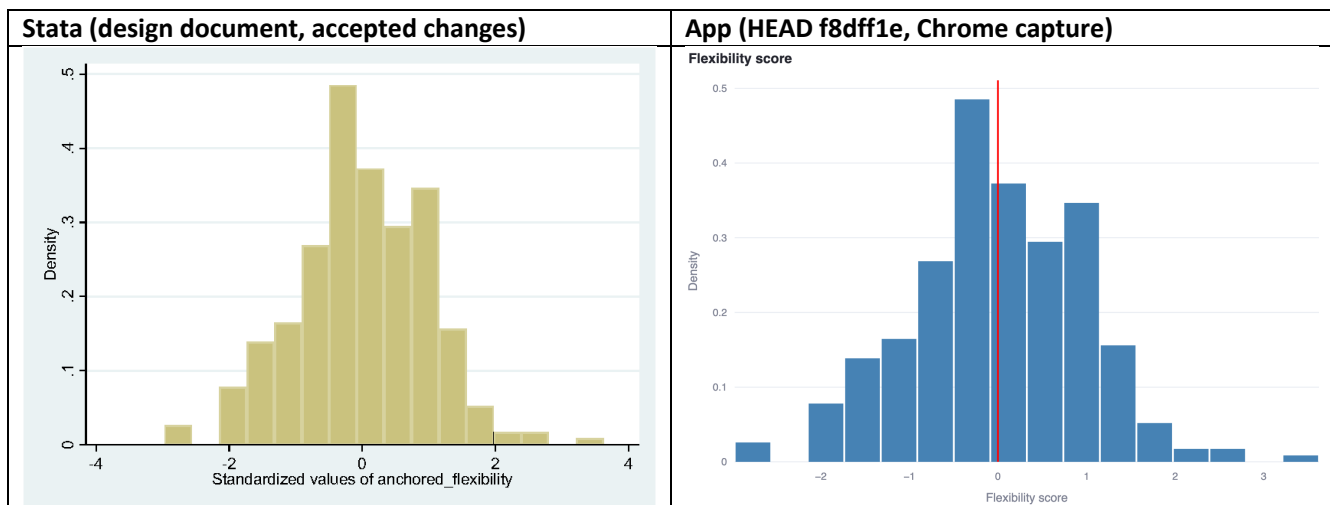
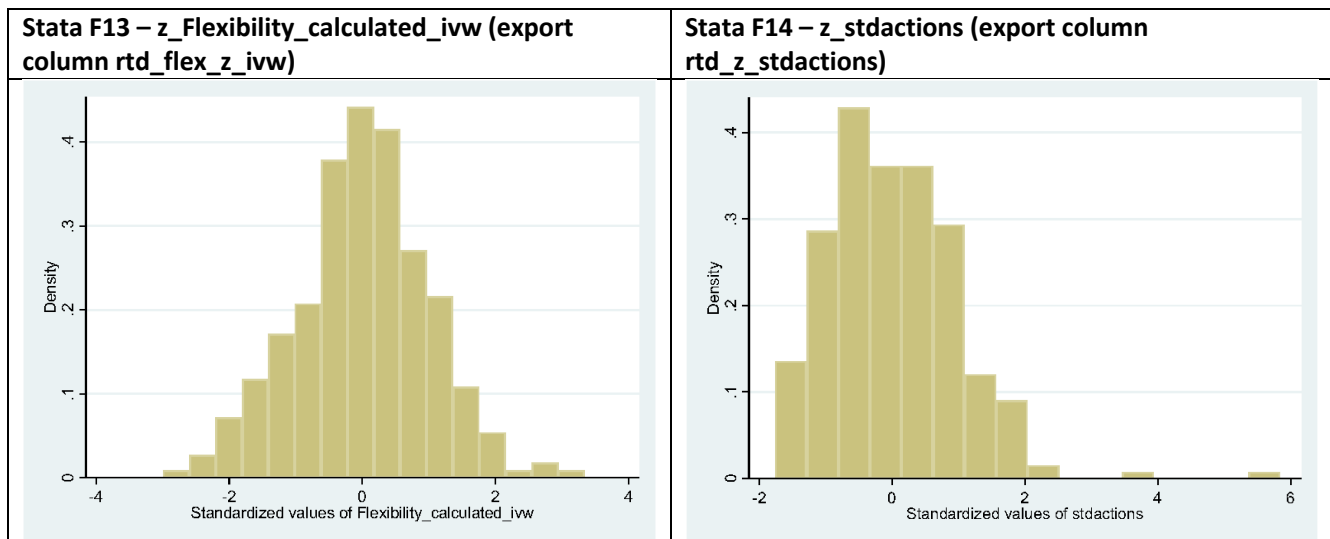
Step 9: Flexibility ranking – F13, F14, F15, F16

Settings as in step 3 ($\beta_4 = 0$, W_OFlex = 0.25; no income term).

```
gen Flexibility_calculated_ivw = 0.0206 * z_extraversionbig5 + 0.0294118 * z_opennessbig5 -
0.04921357 * z_neuroticismbig5 + 0.04339814 * z_agreeable + 0.04811179 * z_conscientiousnessbig5
egen z_Flexibility_calculated_ivw = std(Flexibility_calculated_ivw)
replace z_Flexibility_calculated_ivw = z_Flexibility_calculated_ivw + beta4
histogram z_Flexibility_calculated_ivw
egen z_stdactions = std(stdactions)
histogram z_stdactions
gen anchored_flexibility = 0.25*z_stdactions + 0.75 * z_Flexibility_calculated_ivw
egen z_anchored_flexibility = std(anchored_flexibility)
histogram z_anchored_flexibility
```

F13 and F14 are intermediate variables that the app does not chart; they are compared through the app's export columns (rtd_flex_z_ivw, rtd_z_stdactions). F15 is the app's "Flexibility score" chart.

Variable	Stat a Obs	Stata Mean	Stat a SD	Stata Min	Stata Max	Ap p Obs	App Mean	Ap p SD	App Min	App Max
z_Flexibility_calculated_ivw	280	8.764411e-10	1	-2.994517	3.338073	280	1.268826e-17	1	-2.994517	3.338074
z_stdactions (.dta: z_stdactionsP)	280	5.554674e-09	1	-1.756797	5.835501	280	4.165653e-08	1	-1.756797	5.835501
z_anchored_flexibility	280	3.891598e-10	1	-2.970918	3.623505	280	-2.854859e-17	1	-2.970918	3.623505



```

gen Flexibility_combined15 = floor(1+(5-0.0001)*((z_anchored_flexibility-
min_Flexibility_combined)/(max_Flexibility_combined - min_Flexibility_combined)))
histogram Flexibility_combined15

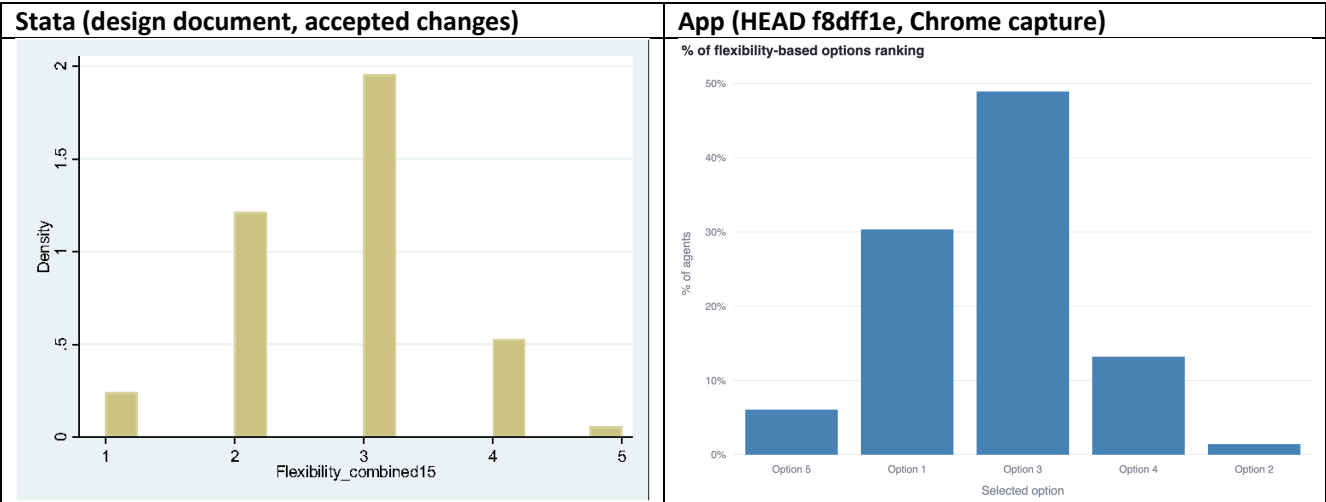
```

Variable	Stata Obs	Stata Mean	Stata SD	Stata Min	Stata Max	App Obs	App Mean	App SD	App Min	App Max
Flexibility_combined15	280	2.735714	0.817468	1	5	280	2.735714	0.817468	1	5

Category	Stata count	Stata %	App count	App %	Equal
segment 1	17	6.1%	17	6.1%	yes
segment 2	85	30.4%	85	30.4%	yes
segment 3	137	48.9%	137	48.9%	yes
segment 4	37	13.2%	37	13.2%	yes
segment 5	4	1.4%	4	1.4%	yes

Category	Stata count	Stata %	App count	App %	Equal
first option 1	85	30.4%	85	30.4%	yes
first option 2	4	1.4%	4	1.4%	yes
first option 3	137	48.9%	137	48.9%	yes

first option 4	37	13.2%	37	13.2%	yes
first option 5	17	6.1%	17	6.1%	yes



The values match exactly: `z_Flexibility_calculated_iwv`, `z_stdactions` and `z_anchored_flexibility` 280/280 (max `|diff|` $\leq 2.2e-07$), `Flexibility_combined15` 280/280, choice lists 280/280. Figures bar by bar – F15: 16 bars vs 16 in the Stata figure, bin edges equal to $1.1e-07$, max bar-height difference $6.9e-10$ density units (0.000 % of the tallest bar); F16: 5 bars vs 5, max bar-height difference 0.000 percentage points. F13 and F14 have no app chart (columns verified 280/280).

Step 10: The σ tables (stochastic component constants)

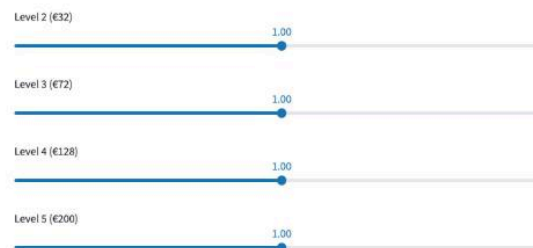
The document derives, per element, $\sigma_{\text{overall}} = (\text{range}/18) \cdot \text{mean}(\text{stdactions})$ and $\sigma_{\text{quintile}(q)} = (\text{range}/18) \cdot \text{mean}(\text{stdactions within allowance group } q)$. `mean(stdactions)` by total allowance from the `.dta`:

Total allowance	<code>mean(stdactions)</code> (.dta)
-----------------	--------------------------------------

12	1.022823
32	1.136698
72	1.310647
128	1.225694
200	1.25449
Total	1.186338

Ranges of the standardised scores (document vs .dta): weighted_loyalty 7.0783045 vs 7.078304; z_WTP_calculated 6.868056 vs 6.868057; z_RT_calculated_hs 5.979747 vs 5.979747; z_anchored_flexibility 6.5944228 vs 6.594423 (float32 storage explains the 7th-digit differences).

Element	Doc σ_{overall}	App base σ	Doc σ_{quintile} 12/32/72/128/200	App σ per level 1–5
TTP	0.395446	0.395446	0.340941 / 0.378899 / 0.436882 / 0.408565 / 0.418163	0.3409412 / 0.3788994 / 0.4368824 / 0.4085648 / 0.4181634
Loyalty	0.4665145336	0.4665145	0.3738315395 / 0.4154515424 / 0.4790281978 / 0.4479787442 / 0.4585033149	0.3738315 / 0.4154515 / 0.4790282 / 0.4479787 / 0.4585033
WTP	0.4526575455	0.4526575	0.3902671709 / 0.4337170115 / 0.5000888363 / 0.4676742829 / 0.4786615699	0.3902672 / 0.433717 / 0.5000888 / 0.4676743 / 0.4786616
Risk-Taking	0.39522204	0.395222	0.33979032 / 0.37762039 / 0.43540774 / 0.40718566 / 0.41675186	0.3397903 / 0.3776204 / 0.4354077 / 0.4071857 / 0.4167519
Flexibility	0.4346228732	0.4346229	0.3747183671 / 0.4164371037 / 0.4801645799 / 0.4490414678 / 0.4595910065	0.3747184 / 0.4164371 / 0.4801646 / 0.4490415 / 0.459591

Base σ and effective σ per budget level:

Budget Level	Options List Length base σ	Options List Length effective σ	Loyalty base σ	Loyalty effective σ	Willingness-to-Pay base σ	Willingness-to-Pay effective σ	Risk-Taking base σ	Risk-Taking effective σ	Flexibility base σ	Flexibility effective σ
Level 1 (€12)	0.340941	0.340941	0.373832	0.373832	0.390267	0.390267	0.33979	0.33979	0.374718	0.374718
Level 2 (€32)	0.378899	0.378899	0.415452	0.415452	0.433717	0.433717	0.37762	0.37762	0.416437	0.416437
Level 3 (€72)	0.436882	0.436882	0.479028	0.479028	0.500089	0.500089	0.435408	0.435408	0.480165	0.480165
Level 4 (€128)	0.408565	0.408565	0.447979	0.447979	0.467674	0.467674	0.407186	0.407186	0.449041	0.449041
Level 5 (€200)	0.418163	0.418163	0.458503	0.458503	0.478662	0.478662	0.416752	0.416752	0.459591	0.459591

Sub-Decision Mechanisms

Two arithmetic slips inside the document itself (the app follows the document literally, as instructed): (a) Risk-Taking σ_{overall} is printed as " $0.332208167 \times 1.1863376 = 0.39522204$ ", but that product is 0.39411104 – the per-level column IS consistent with the factor; (b) the Loyalty σ_{quintile} column (0.3738315395 ...) equals the previous revision's column multiplied once more by 1.0228235 and does not equal $0.393239139 \times \text{mean}(\text{stdactions within group})$ (which would give 0.4022 / 0.4470 / 0.5154 / 0.4820 / 0.4933). TTP, WTP and Flexibility are internally consistent.

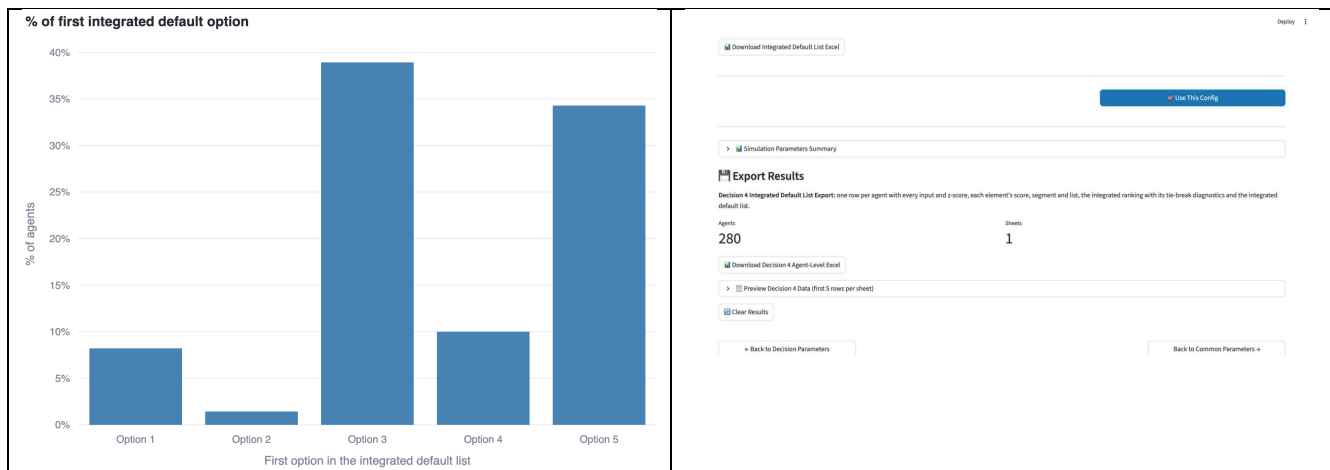
The app constants equal the document's tables (every digit); the two inconsistencies above are in the document, not in the app.

Step 11: Integrated default list (document section 6) – app only

The document specifies Kemeny–Young aggregation of the four ranking lists with the tie-breaking hierarchy Schulze → Copeland → footrule → random, truncated to the options list length and at Option 5. It provides no Stata output for it (its stage shares – Kemeny alone 7.7 %, Copeland 43.3 %, footrule 22.7 %, last resort 26.3 % – come from 100 000 random test cases, not from the 280 participants).

App – first integrated default option (whole-decision run)

App – "Run Integrated Default List Only" (export + Use This Config)



Statistic (Research Baseline, 280, continuous)	App value
Agents with initial ties after Kemeny	98.9% ({'multiple': 215, 'unique': 3, 'unique_with_ties': 62})
Stage that settled the ranking	copeland: 83 (29.6%), footrule: 101 (36.1%), kemeny: 3 (1.1%), random: 93 (33.2%)
Final ranking Kemeny-optimal	100.0%
Truncation rule that bound	{'both': 76, 'length': 100, 'option5': 104}
Integrated list length 0/1/2/3	{'0': 20, '1': 160, '2': 91, '3': 9} (mean 1.32)
First integrated option 1...5 (0 = empty list)	{'0': 20, '1': 23, '2': 4, '3': 109, '4': 28, '5': 96}

App-only step: both output rules hold for 280/280 agents (list length $\leq$ options list length; nothing after Option 5); the integrated-only run and the whole-decision run produce the same frame.

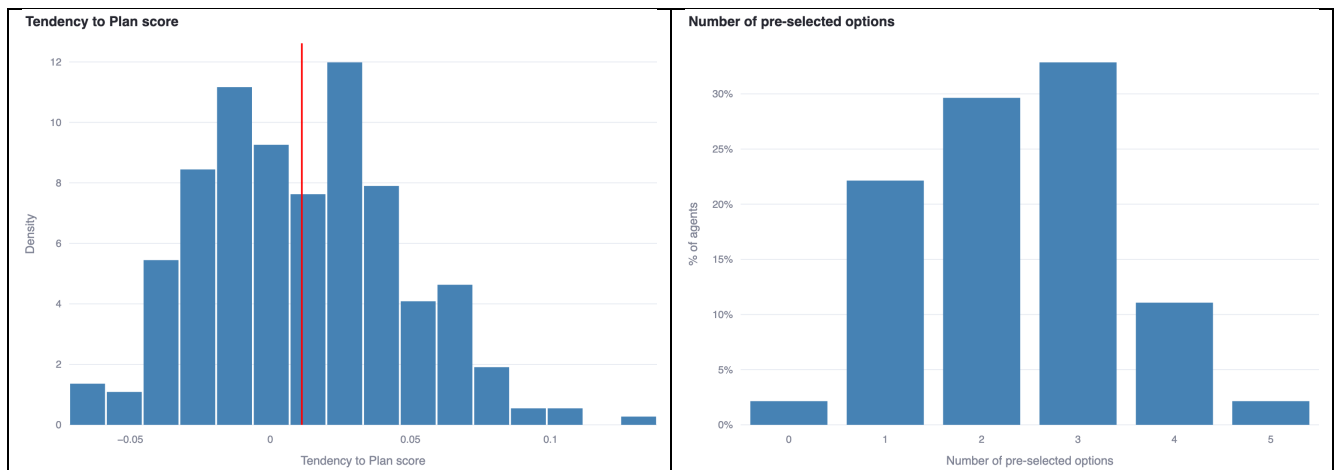
Step 12: Stochastic component – app example (no Stata figure)

Settings: Page 1 Population = Research Specification (the same 280 participants in file order), 280 agents, seed 42; Decision 4 tab: "Use Normal(anchor, σ) draw in Research Specification mode" checked, σ uniform, coefficient 1.00 (σ used: TTP 0.395446, Loyalty 0.4665145, WTP 0.4526575, RT 0.395222, Flexibility 0.4346229); intercepts 0.

The score histograms are those of the deterministic scores (identical to steps 3–9); the draws move the allocations. Options List Length allocation, deterministic vs stochastic:

Options	Deterministic count	%	Stochastic count	% (UI shows 2.1 / 22.1 / 29.6 / 32.9 / 11.1 / 2.1)
0	20	7.1%	6	2.1%
1	92	32.9%	62	22.1%
2	95	33.9%	83	29.6%
3	57	20.4%	92	32.9%
4	14	5.0%	31	11.1%
5	2	0.7%	6	2.1%

App – TTP score (Research Specification, σ on)	App – allocation with the Normal(anchor, σ) draw
---	--



Deploy

presents how many default options each agent pre-selects (0-5)

Min -0.0718 · Max 0.1380



Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

App-only step: the on-screen stochastic allocation equals the engine's for the same seed (UI = engine), and the draws are reproducible by seed.

Step 13: Comparison layouts and Copula

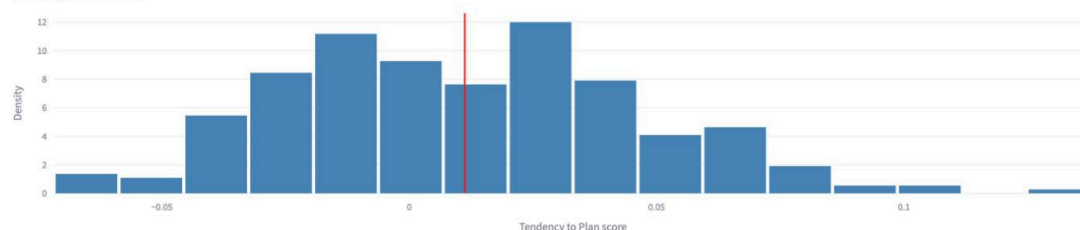
"Compare both" (Decision 4 tab) renders the income-independent elements once and WTP / Risk-Taking / integrated list per income treatment; "Compare all" (Page 1) renders one column per population mode. Each column equals the corresponding single-mode run (verified per agent through the seam).

Options List Length (Tendency to Plan)

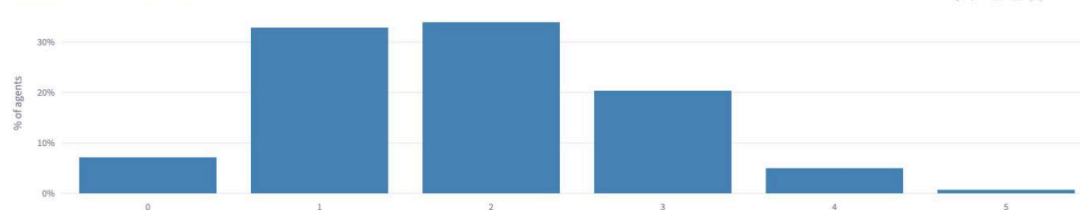
presents how many default options each agent pre-selects (0-5)

Min -0.0718 • Max 0.1380

Tendency to Plan score



Number of pre-selected options



COOPECON AI Agent Simulation

Simulation Results

Decision Results

✓ 4. Rejected Transaction Defaults (Custom Parameters)

This decision was configured with custom parameters

⚙️ Current Settings: Continuous only

🔗 Copula (Synthetic)

📄 Research Specification

📄 Research Baseline

Simulation Overview (Copula, Cont)

Total Agents	Avg. Options List ...	Min Options ...	Max Options ...
280	2.61	0	5

Simulation Overview (Research Spec, Cont)

Total Agents	Avg. Options List ...	Min Options ...	Max Options ...
280	2.35	0	5

Simulation Overview (Research Baseline, Cont)

Total Agents	Avg. Options List ...	Min Options ...	Max Options ...
280	1.85	0	5

🔗 Copula (Synthetic)

📄 Research Specification

📄 Research Baseline

As in the Decision 1 comparison, the Copula column cannot match the Stata values: it is a synthetic population drawn from the fitted copula, never the original 280 participants (here Min -0.0952 / Max 0.1130 for the TTP score vs -0.0718 / 0.1380). Research Specification and Research Baseline carry the original participants in the same file order; with σ off (or σ coefficient 0) Research Specification reproduces Research Baseline agent for agent (verified on every rtd_* column and list).

Consistent: the comparison grids reproduce the single-mode values; Copula differs by construction.

Appendix A: Bar-by-bar check of every figure pair

For each Stata figure the document's bin count was established from the figure itself (bar heights measured in the image and matched against np.histogram of the .dta column at 14...18 bins: all eleven continuous histograms match

16 bins with a maximum relative bar error $\leq 0.5\%$ – F14 2.1 % because of its very small bars – while 17 bins give 19–40 %; the five discrete figures match the value counts). The app chart's bars were then read from the running app's Plotly figure objects (bin edges, densities / percentages) and compared with the same np.histogram / value counts. "Max bar difference" is the largest absolute difference in bar height over all bars; for histograms it is also given relative to the tallest bar.

Stata figure	.dta column	App chart	Bars Stata / app	Max edge diff	Max bar difference	Same
F1	weighted_ttp	Tendency to Plan score	16 / 16	7.7e-09	3.6e-08 (0.000 % of tallest bar)	yes
F2	choice_length_deterministic	Number of pre-selected options	6 / 6	–	0.000 pp	yes
F3	weighted_loyalty	Loyalty score	16 / 16	7.2e-08	2.6e-09 (0.000 % of tallest bar)	yes
F4	weighted_loyalty15	% of loyalty-based options ranking	5 / 5	–	0.000 pp	yes
F6	z_WTP_calculated	Willingness-to-Pay score (Continuous only)	16 / 16	3.9e-06	5.8e-07 (0.000 % of tallest bar)	yes
F8	WTP_calculated15	% of willingness-to-pay-based options ranking (Continuous only)	5 / 5	–	0.000 pp	yes
F7	z_WTP_categorical (stored .dta column, superseded formula)	Willingness-to-Pay score (Categorical only)	16 / 16	1.7e-02	1.6e-01 (24.686 % of tallest bar)	NO
F7 → floor rule	segments of the stored z_WTP_categorical	% of willingness-to-pay-based options ranking (Categorical only)	5 / 5	–	0.357 pp	NO
F9	z_RT_calculated_hs	Risk-Taking score (Continuous only)	16 / 16	7.4e-08	1.3e-09 (0.000 % of tallest bar)	yes
F12	RT_calculated15	% of risk-taking-based options ranking (Continuous only)	5 / 5	–	0.000 pp	yes
F11	z_RT_categorical	Risk-Taking score (Categorical only)	16 / 16	1.5e-07	1.3e-08 (0.000 % of tallest bar)	yes
F11 → floor rule	segments of z_RT_categorical	% of risk-taking-based options ranking (Categorical only)	5 / 5	–	0.000 pp	yes
F15	z_anchored_flexibility	Flexibility score	16 / 16	1.1e-07	6.9e-10 (0.000 % of tallest bar)	yes
F16	Flexibility_combined15	% of flexibility-based options ranking	5 / 5	–	0.000 pp	yes
F5, F10	weighted_WTP_categorical, weighted_RT_categorical (raw scale)	no app chart (the app charts the standardised score; same histogram up to the x-scale)	–	–	–	n/a

F13, F14	z_Flexibility_calculated_ivw, z_stdactionsP	no app chart (export columns rtd_flex_z_ivw, rtd_z_stdactions: 280/280)	–	–	–	n/a
----------	--	---	---	---	---	-----

Appendix B: Per-participant verification against the .dta (controller script, seam path)

Research Baseline, 280 agents, seed 42, intercepts 0, W_OFlex 0.25, σ off; rows joined on Participant ID.

Stata column	App column	Participants matching	max diff
weighted_ttp	rtd_weighted_ttp	280/280	1.8e-08
choice_length_deterministic	rtd_choice_length	280/280	0
weighted_loyalty	rtd_loyalty_z	280/280	3.2e-07
weighted_loyalty15	rtd_loyalty_segment	280/280	0
choice1...5_loyalty_deterministic	rtd_loyalty_ranking	280/280	0
z_WTP_calculated (continuous)	rtd_wtp_z	280/280	8.1e-06
WTP_calculated15	rtd_wtp_segment	280/280	0
choice1...5_WTP_deterministic	rtd_wtp_ranking	280/280	0
z_WTP_categorical (stored .dta column)	rtd_wtp_z (categorical)	0/280 – superseded formula in the .dta (step 6); app = document code to 9e-09	6.2e-02
floor rule on z_WTP_categorical (.dta)	rtd_wtp_segment (categorical)	279/280	–
z_RT_calculated_hs (continuous)	rtd_rt_z	280/280	1.7e-07
RT_calculated15	rtd_rt_segment	280/280	0
choice1...5_RT_deterministic	rtd_rt_ranking	280/280	0
z_RT_categorical	rtd_rt_z (categorical)	280/280	2.0e-07
floor rule on z_RT_categorical	rtd_rt_segment (categorical)	280/280	–
z_Flexibility_calculated_ivw	rtd_flex_z_ivw	280/280	1.8e-07
z_stdactionsP	rtd_z_stdactions	280/280	1.6e-07
z_anchored_flexibility	rtd_flex_z	280/280	2.2e-07
Flexibility_combined15	rtd_flex_segment	280/280	0
choice1...5_flex_deterministic	rtd_flex_ranking	280/280	0

Appendix C: Document / data errors found

- The stored column weighted_WTP_categorical in Stata_File_Decision4_050926.dta (and therefore z_WTP_categorical and figure F7) was produced with a superseded formula (see step 6); the document's printed cond() code is what the app implements.
- The document's histograms use Stata's default bin count, which TRUNCATES $\min(\sqrt{N}, 10 \cdot \log_{10} N) = 16.73$ to 16 bins at $N = 280$ (verified bar by bar on all eleven continuous figures). The app rounded to 17 bins until HEAD e41968d; the earlier revision of this comparison stated "identical 17-bin histograms" on the basis of summary statistics, which was wrong.
- Risk-Taking σ_{overall} : the document prints $0.332208167 \times 1.1863376 = 0.39522204$; the product is 0.39411104.
- Loyalty σ_{quintile} column: the document's five per-allowance values are the previous revision's column multiplied by 1.0228235 again and do not follow from the stated factor 0.393239139.
- Flexibility: the document lists "histogram z_Flexibility_calculated_hs" and Risk-Taking "histogram z_RT_calculated_ivw" without the corresponding figures (the IVW/H&S alternatives that were not retained).
- The document still contains tracked changes (deleted previous-revision figures and choice mappings); this comparison uses the accepted-changes text and figures.